

Disclosed Intelligence: A Large-Sample Measurement of AI Disclosure in U.S. Investment Adviser Fiduciary Filings

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Abstract

Industry surveys report that a majority of registered investment advisers (RIAs) now use artificial intelligence (AI), but what advisers commit to in their legally binding fiduciary disclosures is largely unmeasured. We provide the first large-sample, classifier-based measurement of AI disclosure in U.S. Form ADV Part 2A brochures. From the SEC Form ADV Part 1 structured data (16,223 active SEC-registered advisers reporting regulatory AUM), we draw a stratified random sample of 400 firms (adviser type $\times$ assets-under-management quartile) and retrieve each firm's current brochure from the Investment Adviser Public Disclosure system; brochures were classified for 388 firms (the remaining 12, all in the wealth/retail strata, were brochure-exempt or yielded no usable text). A five-label typology—AI in the investment process, in operations/client service, as a disclosed risk factor, explicit non-use, and named vendor—is applied by a large language model, checked for same-family reproducibility and validated by an independent, blinded re-coding of 180 brochures by a different model family (any-use $\kappa = 0.65$, recall 0.88). We report three results with design-based confidence intervals. First, disclosed AI use is low and steeply stratified: 23.7% of brochures (95% CI 19.7–28.2) disclose any use, increasing strongly with adviser size from about 9% among small advisers to 30% (wealth/retail) and 56% (private-fund) among the largest, with strong trend and adviser-type effects (logistic regression, both $p < 0.001$). A survey-weighted estimate that reweights the strata to the brochure-filing population is 22.7% (95% CI 18.7–26.6), close to the design estimate, so disclosed use is robust to the sampling design. Second, the most common disclosure mode is a risk warning (27.6%), not a claim of use. Third, potential AI-washing exposure—language resembling conduct the SEC has charged—is rare in brochures (five documents, 1.3%; on adjudication none is an unsubstantiated promotional claim of the charged type) and, in a matched 283-firm comparison, no larger in current marketing; firms disclose AI use more in the lawyered brochure (23.0%) than on their websites (5.7%). We frame the survey-versus-disclosure comparison as a benchmark divergence between differently defined measures, not a validated nondisclosure rate, and we detail reliability, weighting, and scope limitations.

Keywords: artificial intelligence, generative AI, investment advisers, Form ADV, fiduciary disclosure, text classification, large language models, AI-washing

1 Introduction

Generative artificial intelligence has moved from research demonstration to a claimed business capability across financial services within a few years. Industry surveys report that a majority of registered investment advisers (RIAs) use or are piloting AI, and the Charles Schwab 2026 RIA and AI research study places adoption at roughly 63% of surveyed firms. Adoption surveys capture what advisers report to pollsters; they do not capture what advisers commit to in the documents that carry legal weight—the Form ADV Part 2A “brochure” that every SEC-registered adviser must deliver to clients and that anchors the adviser's fiduciary and anti-fraud obligations under the Investment Advisers Act (U.S. SEC, 2019). Whether, how, and where advisers disclose AI in these filings has not been measured at industry scale.

This matters because disclosure is the mechanism through which the fiduciary standard is operationalized, because the SEC brought its first two “AI-washing” enforcement actions in March 2024 (against Delphia and Global Predictions) for misrepresenting AI capabilities, and because a reliable, reproducible instrument for AI claims in regulatory text is a public good for a 16,000-firm population no team can read by hand. Our aims are deliberately measurement-first: to build and validate such an instrument, to describe the level and distribution of disclosed AI use, and to compare disclosed use against the surveyed-use benchmark and against firms’ own marketing.

We draw a stratified random sample of 400 advisers from the SEC Form ADV Part 1 structured data, stratifying on adviser type (private-fund versus wealth/retail) and AUM quartile; retrieve each firm’s current brochure; classify every brochure with a preregistered five-label typology using a large language model; and quantify uncertainty and reliability. Three findings emerge, each reported with design-based confidence intervals. Disclosed any-use is 23.7% (95% CI 19.7–28.2), increases strongly with size, and is higher for private-fund advisers at every size (trend and type effects both $p < 0.001$). The most common disclosure mode is risk-framing (27.6%), not a claim of use. Potential AI-washing exposure is rare in brochures (1.3%) and, in a matched comparison, no larger in current marketing; the fiduciary brochure is, if anything, the more AI-forward venue.

On the survey-versus-disclosure comparison. We report a divergence of about 39 percentage points between disclosed any-use (23.7%) and the 63% surveyed-use benchmark. We are explicit that this is a benchmark divergence between two differently defined measures—different populations (all SEC registrants versus Schwab-custodied survey respondents), different dates, adviser-level versus firm-level reporting, and experimental or administrative “use” versus material, disclosable use—and not a validated within-firm nondisclosure rate. It is informative as a contextual benchmark divergence, but the point estimate should not be read as the fraction of AI adopters who fail to disclose—the survey could overstate organizational adoption, and brochures could capture algorithmic practices respondents would not call “AI.” Establishing that quantity would require matched adoption information for the sampled firms, which we flag as the natural next step.

Our contributions are: (i) a preregistered, reproducible typology for AI claims in adviser filings with reported same-family reproducibility and independent cross-family validation; (ii) the first large-sample estimate of disclosed AI use and its size/type gradient, with design-based inference; and (iii) an enforcement-anchored exposure screen and a venue comparison. Section 2 situates the study; Section 3 gives data, methods, and inference; Section 4 reports results; Section 5 discusses interpretation, regulation, and limitations.

2 Institutional Background and Related Literature

2.1 Form ADV and the fiduciary brochure

Every SEC-registered adviser files Form ADV. Part 1 is structured, machine-readable firm data (legal name, CRD identifier, regulatory AUM, client counts, private-fund activity, disciplinary history), published by the SEC as bulk data. Part 2A, the brochure, is a narrative plain-English document describing the adviser’s business, strategies, fees, conflicts, and material risks; it must be delivered to clients and is the primary written expression of the adviser’s disclosure obligations. Because the brochure describes how an adviser actually manages money, it is where disclosed AI use should appear, and where misstatements about AI carry the clearest legal consequence.

2.2 AI-washing enforcement

On 18 March 2024 the SEC settled charges against two advisers for false or misleading AI statements. Delphia (USA) Inc. claimed it used “machine learning to analyze the collective data” in a “predictive algorithmic model” it had not developed (US \$225,000 penalty); Global Predictions Inc. billed itself the “first regulated AI financial advisor” with “[e]xpert AI-driven forecasts” it could not substantiate (US \$175,000) (U.S. SEC, 2024a,b). These orders establish that promotional AI claims are actionable and provide a textual fingerprint of charged conduct against which live disclosures can be screened.

2.3 Related literature

Our design template is the machine-classification-of-filings tradition: Abis (2022) classifies mutual-fund prospectus text to separate quantitative from discretionary managers and links it to fund behavior. We adapt that move—text as the measurement instrument for a firm attribute—to AI disclosure. Babina et al. (2024) measure AI adoption from job postings and relate it to firm growth; the divergence between adoption so measured and fiduciary disclosure is our object. Methodologically we build on regulatory-filings textual analysis (Loughran and McDonald, 2011, 2016; Gentzkow et al., 2019) and on validation practice for computational text measurement (Grimmer and Stewart, 2013). We use large language models as annotators, a rapidly maturing method whose reliability must be established against human coding (Gilardi et al., 2023; Ziems et al., 2024). The broader economics of AI adoption (Acemoglu et al., 2022; Eisfeldt et al., 2023; Babina et al., 2024) and of generative-AI productivity (Noy and Zhang, 2023; Brynjolfsson et al., 2025) motivate why disclosure is worth measuring; disclosure theory (Healy and Palepu, 2001) frames why firms may withhold it; and work on strategic disclosure when machines read filings (Cao et al., 2024) is a close conceptual neighbor. To our knowledge this is the first large-sample, classifier-based measurement of AI disclosure in adviser fiduciary filings.

3 Data and Methods

3.1 Sampling frame and stratified sample

The frame is the SEC Form ADV Part 1 structured data (adviser base table). We keep each firm’s most recent filing and restrict to firms active through 2024. From 27,679 unique CRD identifiers filing between 2011 and 2024, this yields 16,709 active registrants, of which 16,223 report regulatory AUM and constitute the sampling and weighting universe. We draw a stratified random sample of 400 firms with a fixed seed, crossing adviser type—private-fund (Item 7.B) versus wealth/retail—with AUM quartile (Item 5.F.(2)(c) total regulatory AUM), eight cells of 50. The sample is frozen before any brochure is read. Equal allocation deliberately over-samples large advisers relative to the small-firm-dominated population; we therefore report both design-based (unweighted) and survey-weighted estimates (Section 3.7).

3.2 Brochure retrieval and classified denominators

For each firm we obtain the current Part 2A brochure version identifier from the IAPD firm API, retrieve the brochure PDF, and extract its text. Text was classified for 388 of 400 firms. The 12 firms without a classified brochure are all in the wealth/retail strata: 11 returned an IAPD Part-2-brochure-exempt flag and one retrieved document yielded no usable text. Because these fall in the wealth cells, wealth-stratum denominators are 46–48 rather than 50, which we carry through all per-cell estimates. The estimand is disclosure among advisers that file a Part 2A brochure; brochure-exempt firms have no brochure and are excluded from the disclosure estimand rather than counted as zero-disclosers, and the single unretrievable document is treated as missing. Because exemptions are rare (11 of 400), an alternative that counted exempt firms as non-disclosers would lower the prevalence by at most about three percentage points and would not

change any conclusion. Retrieval respected the host’s fair-access expectations; raw documents were persisted with content hashes.

3.3 The claim typology

The instrument is a five-label, multi-label typology, fixed before analysis: (a) AI in the investment process; (b) AI in operations/client service; (c) AI as a disclosed risk factor; (d) explicit prohibition/non-use; (e) a named AI vendor/product invoked as an AI capability. Derived indicators are any-use ($a \cup b \cup e$) and any-mention ($a \cup b \cup c \cup d \cup e$). Because label (e) records only that a vendor is named—not necessarily that the adviser materially uses it—we report any-use both including and excluding (e). A label is assigned only when a verbatim passage supports it. The full rubric, prompts, keyword-window rules, model snapshot and date, and decision rules are deposited (Section 6).

3.4 LLM classification and reliability

Classification is tooling; the object of study is the industry. AI-relevant passages are extracted by keyword windowing and submitted with the frozen rubric to a large language model (OpenAI gpt-4o, fixed snapshot, temperature 0) that returns structured labels. We assess reliability at two levels. Inter-model reproducibility: a random 60-brochure subsample is re-classified by a second model from the same family (gpt-4o-mini) and we report per-label percent agreement and Cohen’s κ (Cohen, 1960); agreement between two same-family models establishes reproducibility, not accuracy. Independent cross-family validation: a blinded 180-brochure set—all model-positive brochures plus a random sample of model-negative brochures (seed 42)—is re-coded from the brochure text and the frozen rubric alone by an independent rater from a different model family (Anthropic Claude), with no access to the primary labels. Against this independent re-coding we report, for the primary classifier, precision, recall, F1, confusion counts, Cohen’s κ , and prevalence-and-bias-adjusted agreement (PABAK; Byrt et al., 1993) per label and for the derived any-use indicator. Because both raters are language models, this measures cross-family, cross-instrument agreement against a common rubric—stronger than the same-family check (larger sample, fully blinded, independent family) but not a substitute for a panel of human domain experts; we therefore enumerate the full residual disagreement set so a targeted human audit can adjudicate the boundary cases (Supplementary Material S2). The coding sheet, the independent codes, and the scoring code are provided so the step is fully auditable. Total metered classification cost was under two U.S. dollars.

3.5 The exposure screen

From the text of the Delphia and Global Predictions orders we distill seven promotional-claim patterns (F1–F7: predictive-power, first-mover/identity, edge/alpha, AI-manages-portfolios, learns-from-data, quantified-capability, and AI-as-headline-differentiator). A document is flagged if it makes a claim materially similar to any pattern. We deliberately term this a potential exposure-language screen: it measures textual similarity to charged conduct, not misconduct; the fingerprint is distilled from only two cases; positives are few; and every positive is human-adjudicated with the passage quoted (Supplementary Material S1). All results are aggregate and no firm is identified.

3.6 Marketing corpus, venue comparison, and missingness

To test venue substitution, we assemble a current marketing corpus for the sampled firms. Each firm’s website is resolved from its brochure; the homepage and principal descriptive pages (about, approach, technology) are crawled (one request per second per host, up to eight pages, main-text extraction) and classified with the same typology and exposure screen. Usable marketing text was obtained for 283 firms. We characterize selection explicitly (Section 4.5): we compare firms with and without usable marketing text

on adviser type, AUM quartile, and regulatory AUM. The comparison is limited to current main-site pages; JavaScript-rendered content, press releases, social media, and historical campaigns are out of scope, a limitation we foreground because the enforcement claims themselves appeared in such channels.

3.7 Inference and weighting

We report Wilson 95% confidence intervals for every prevalence, per label and per stratum. We test the AUM gradient with a linear-by-linear trend test within each adviser type and overall, and estimate a firm-level logistic regression of any-use on AUM quartile (ordinal) and adviser type. The venue difference is tested with a paired bootstrap. For population inference we compute a survey-weighted any-use prevalence over the estimand’s population—brochure-filing advisers. Each stratum’s exact population count (from applying the identical stratification to the full frame) is deflated by the stratum’s observed brochure-filing rate to estimate the brochure-filing population, and each stratum’s sample prevalence is weighted by its share of that universe, with a stratified design-based variance. Using the unadjusted all-AUM-reporting universe (16,223 firms) instead yields essentially the same value (22.3% vs. 22.7%), confirming that the small brochure exemptions do not affect the estimate. Weighted and unweighted estimates are reported side by side (Table 3).

4 Results

4.1 Typology distribution

Table 1 reports each label’s prevalence with Wilson 95% CIs across the 388 classified brochures. Disclosed any-use ($a \cup b \cup e$) is 23.7% (19.7–28.2); excluding the named-vendor label, any-use of ($a \cup b$) is 20.9%. Any AI mention is 31.7% (27.3–36.5). The most common single label is AI as a disclosed risk factor (27.6%), exceeding investment-process use (18.3%); advisers more often warn about AI than claim to use it. Explicit non-use (d) is essentially absent.

Label	k / n	Prevalence (95% CI)
(a) AI in investment process	71 / 388	18.3% (14.8–22.5)
(b) AI in operations / client service	26 / 388	6.7% (4.6–9.6)
(c) AI as a disclosed risk factor	107 / 388	27.6% (23.4–32.2)
(d) Explicit prohibition / non-use	8 / 388	2.1% (1.0–4.0)
(e) Named AI vendor / product	21 / 388	5.4% (3.6–8.1)
Any use ($a \cup b \cup e$)	92 / 388	23.7% (19.7–28.2)
Any mention (a–e)	123 / 388	31.7% (27.3–36.5)

Table 1. Typology prevalence with Wilson 95% confidence intervals (gpt-4o classifier; n = 388 classified brochures). Labels are multi-label.

4.2 Size and type gradient

Table 2 shows disclosed any-use by stratum with denominators and CIs. Any-use rises monotonically with AUM among private-fund advisers (12% → 24% → 34% → 56%) and broadly increases among wealth/retail advisers (8.7%, 8.3%, 14.6%, 30.4%; the Q1–Q2 difference is within sampling error), and private-fund advisers disclose more at every quartile. A linear-by-linear trend test confirms the gradient (private fund $z = 4.82$, $p < 10^{-5}$; wealth/retail $z = 3.00$, $p = 0.003$; overall $z = 5.57$, $p < 10^{-7}$). In a firm-level logistic regression of any-use on AUM quartile (ordinal) and adviser type, both predictors are significant (AUM

quartile: coefficient 0.67, SE 0.12, $p < 0.001$; private-fund: coefficient 1.01, SE 0.27, $p < 0.001$), and private-fund advisers disclose at 31.5% versus 15.4% for wealth/retail ($\chi^2 p = 0.0003$).

Adviser type	Q1	Q2	Q3	Q4
Private fund	12% (6–24) [6/50]	24% (14–37) [12/50]	34% (22–48) [17/50]	56% (42–69) [28/50]
Wealth / retail	8.7% (3–20) [4/46]	8.3% (3–20) [4/48]	14.6% (7–27) [7/48]	30.4% (19–45) [14/46]

Table 2. Disclosed any-use by adviser type and AUM quartile: prevalence (Wilson 95% CI) [count/denominator]. Wealth/retail denominators are below 50 because the 12 unclassified firms fall in these strata.

4.3 The survey-versus-disclosure divergence

Table 3 reports the eight-stratum brochure-filing population counts, sample sizes, disclosure rates, and sampling weights. Reweighting the strata to the brochure-filing population yields a survey-weighted any-use prevalence of 22.7% (95% CI 18.7–26.6), close to the design-based estimate of 23.7% (19.7–28.2); the unadjusted all-firms universe gives 22.3%. Wealth/retail advisers collectively account for about 64% of the weighting universe and have a population-weighted disclosure rate well below that of private-fund advisers (their two smallest-AUM strata disclose about 8%, though their largest discloses 30%); the large private-fund strata disclose enough (up to 56%) that the weighted estimate remains near the design value. Disclosed use is therefore robust to the sampling design rather than an artifact of over-sampling large firms. Against the 63% surveyed-use benchmark the divergence is about 40 percentage points weighted and 39 unweighted. As stated in Section 1 and Section 5.1, this is a divergence between differently defined measures, not a within-firm nondisclosure rate; the near-equality of the weighted and unweighted estimates simply shows the divergence does not depend on the design.

Stratum	Filing pop. N	Class. n	Any-use	Weight
Private fund · Q1	769	50	12.0%	0.049
Private fund · Q2	1,060	50	24.0%	0.068
Private fund · Q3	1,604	50	34.0%	0.103
Private fund · Q4	2,449	50	56.0%	0.157
Wealth/retail · Q1	3,271	46	8.7%	0.210
Wealth/retail · Q2	2,790	48	8.3%	0.179
Wealth/retail · Q3	2,268	48	14.6%	0.145
Wealth/retail · Q4	1,397	46	30.4%	0.090
Total (weighted)	15,606	388	22.7%	1.000

Table 3. Brochure-filing population counts (exact stratum population deflated by the stratum brochure-filing rate), classified sample sizes, disclosure rates, and sampling weights. The survey-weighted any-use prevalence is 22.7% (95% CI 18.7–26.6) versus the design-based 23.7%; the unadjusted all-firms universe (16,223) gives 22.3%.

4.4 Reliability

Reliability is assessed at two levels. Same-family inter-model agreement on the 60-brochure subsample is near-perfect for the risk-factor label ($\kappa = 0.95$), named-vendor ($\kappa = 1.00$), and any-mention ($\kappa = 0.86$), and moderate for the finer use distinctions (investment-process $\kappa = 0.61$; operations $\kappa = 0.66$; any-use composite $\kappa = 0.69$); the prohibition label has near-zero prevalence, so its κ is undefined. The stronger test is the independent cross-family validation of Table 4: all 180 blinded brochures were re-coded by a different

model family with no access to the primary labels. It confirms the reliable constructs on a larger, blinded, cross-family sample—risk factor $\kappa = 0.887$ ($F1 = 0.951$) and named vendor recall 1.000—and it quantifies the accuracy of the headline estimand: the any-use indicator reaches $\kappa = 0.645$ with recall 0.875 and precision 0.761, so the classifier recovers seven of every eight independently-confirmed adopters. The finer investment-process label is the softest ($\kappa = 0.573$), and the full disagreement analysis (Supplementary Material S2) shows why: the classifier over-calls investment-process AI on reserve-the-right-to-use language, third-party or execution algorithms, and rule-based robo-advice, and it over-calls named-vendor when a product (e.g., ChatGPT) is named only illustratively; in the opposite direction it misses some firms' own quantitative-model investing. These two error directions partly offset, so a first-order bias correction moves disclosed any-use from 23.7% to about 20.6%, within the design-based confidence interval (19.7–28.2). Correcting the known over-calls lowers disclosed use and therefore widens, rather than narrows, the wedge against surveyed adoption; we accordingly report named-vendor and any-use as the load-bearing constructs and read the fine investment-process label as approximate.

Label / indicator	Prec.	Rec.	F1	% agree	κ	PABAK
(a) investment process	0.690	0.778	0.731	80.0%	0.573	0.600
(b) operations	0.808	0.677	0.737	91.7%	0.688	0.833
(c) risk factor	0.916	0.990	0.951	94.4%	0.887	0.889
(d) explicit non-use	0.875	0.467	0.609	95.0%	0.585	0.900
(e) named vendor	0.524	1.000	0.688	94.4%	0.660	0.889
Any use ($a \cup b \cup e$)	0.761	0.875	0.814	82.2%	0.645	0.644

Table 4. Independent cross-family validation on the blinded 180-brochure set (all model-positive brochures plus a random model-negative sample). Metrics are for the primary classifier (gpt-4o) with an independent blind re-coding by a different model family as reference; $n = 180$. Both raters are language models, so this bounds cross-instrument error and is not a human gold standard (Supplementary Material S2). Same-family reproducibility (gpt-4o vs. gpt-4o-mini, 60 brochures) is reported in the text.

4.5 Exposure and the venue comparison

Applying the F1–F7 screen to the brochures, potential exposure language is rare: 5 of 388 brochures (1.3%, 95% CI 0.6–3.0). Given so few positives, each was adjudicated against the charged-conduct standard with the passage quoted (Supplementary Material S1). Two are false positives—a systematic strategy's trading mechanism and an automated managed-account program described factually—and three are substantiated, risk-disclosed descriptions of genuinely operated AI/ML investment methods (the firms name platforms or data vendors and disclose model and predictive-analysis risk). None is an unsubstantiated promotional claim of the type charged in Delphia or Global Predictions; adjudicated charged-conduct-type exposure is therefore 0 of 388 (at most one borderline but substantiated predictive-performance claim). The raw screen rate overstates exposure because the lexical pattern also matches factual quantitative, robo, and machine-learning disclosure—illustrating why the screen is reported as a prefilter requiring adjudication, not a misconduct measure.

For the 283 firms with both a classified brochure and usable marketing text, Table 5 reports the venue comparison. Marketing exposure equals brochure exposure in this matched set (0.7% each; paired difference 0.000, bootstrap 95% CI -0.014 to $+0.014$). Disclosed any-use is markedly higher in the brochure (23.0%) than in marketing (5.7%): 58 firms disclose AI use in the brochure but not on their website, versus 9 in the reverse direction. Contrary to the intuition motivating AI-washing concern, the fiduciary brochure is the more AI-forward venue. A missingness analysis limits selection concern: firms with and without usable marketing text do not differ significantly in adviser type (χ^2 $p = 0.83$), AUM quartile (χ^2 $p = 0.50$), or

regulatory AUM (median US \$419M vs. US \$357M; Mann–Whitney $p = 0.46$). We nonetheless restrict this finding to the matched sample and to current main-site pages.

Measure (n = 283)	Brochure	Marketing	Marketing – Brochure
Exposure-language screen (F1–F7)	0.7%	0.7%	0.000 (–0.014, +0.014)
Any-use disclosure (a $\cup$ b $\cup$ e)	23.0%	5.7%	–17.3 pp

Table 5. Venue comparison for firms with both venues (n = 283). The difference column is Marketing minus Brochure; the bracketed figures are a paired bootstrap 95% CI.

5 Discussion

5.1 What the divergence is, and is not

The survey-versus-disclosure divergence combines several distinct differences: firm populations (all SEC registrants versus Schwab-custodied respondents), observation dates, adviser-level versus firm-level reporting, experimental or individual use versus material organizational use, and general technology use versus a disclosure obligation. It is therefore an informative benchmark divergence, not a validated adoption–disclosure gap, and we do not claim a single nondisclosure percentage. The paper’s primary contributions are the measured level and distribution of disclosure and the instrument that produces them; the divergence is context. A design that obtained matched adoption for the sampled firms—by survey linkage or by coding adoption signals directly—would convert the divergence into a within-firm gap and is the natural extension.

5.2 Interpreting the level and gradient

Disclosed use is low and steeply size-graded. We observe what firms describe, not what they do: larger private-fund advisers are substantially more likely to describe AI-related investment methods in their brochures, and the long tail of small advisers describes none. That the dominant mode is risk-framing rather than a claim of use is consistent with two non-exclusive interpretations—post-enforcement caution and materiality judgments (many productivity or back-office uses may be immaterial to a client’s investment decision and hence not brochure items)—but the cross-sectional design cannot establish that enforcement caused the caution; that would require longitudinal evidence. The significant type effect—private-fund advisers describe AI more at every size—fits their more technical brochures and audiences.

5.3 Why the lawyered document is the louder venue

The brochure discloses AI use more often than firms’ own marketing. The two genres differ: the brochure exhaustively itemizes methods, including algorithmic and quantitative ones and their risks, whereas marketing homepages are short and benefit-oriented; and post-enforcement incentives plausibly cut against promotional AI claims in every public venue (an interpretation the cross-sectional design cannot establish causally). The implication for AI-washing policy is that, in current main-site marketing, the feared venue substitution is not evident—exposure language is rare everywhere we look.

5.4 Regulatory implications

The reliable constructs (any-mention, risk-factor, named-vendor) can be computed across the full population to triage examinations, and the size gradient predicts where investment-process AI concentrates. The exposure screen is a low-base-rate filter for language resembling charged conduct; because such language is rare in brochures, a full deployment should also cover marketing, press, and social text and

historical snapshots. Nothing here identifies misconduct: the screen measures textual similarity, not a legal conclusion, and only aggregates are reported.

5.5 Limitations

First, construct and comparability: the survey benchmark and the disclosure measure are not equivalent (Section 5.1). Second, scope: this is a cross-section of current brochures, not a panel; a short forward panel is the natural extension for diffusion. Third, reliability: same-family inter-model agreement establishes reproducibility, not accuracy, and the independent cross-family validation (Section 4.4, Supplementary Material S2) is itself between two language models rather than against a panel of human experts, so it bounds cross-instrument error but does not certify human-level accuracy; the fine investment-process label is therefore reported as approximate ($\kappa = 0.57$) while any-use ($\kappa = 0.65$) and named-vendor carry the argument, and the enumerated disagreement set is provided for a targeted human audit. Fourth, weighting: the design over-samples large firms, but the design-based (23.7%) and survey-weighted (22.7%) estimates are close, so the reported level is not an artifact of the allocation. Fifth, the exposure screen rests on two enforcement cases and few positives; it is a potential-exposure-language screen with adjudicated positives, not a misconduct measure. Sixth, the venue comparison covers current main-site marketing for 283 of 400 firms; although missingness is not significantly related to size or type, JavaScript-rendered, press, social, and historical content are unobserved.

6 Conclusion

We provide the first large-sample, classifier-based measurement of AI disclosure in U.S. investment-adviser fiduciary filings. Disclosed AI use is low (23.7%; 95% CI 19.7–28.2), steeply graded by size and type, and dominated by risk-framing; it diverges widely from surveyed adoption, a divergence we interpret carefully rather than as a nondisclosure rate. Potential AI-washing exposure is rare in brochures and no larger in current marketing. The instrument is reproducible on the constructs that matter, is validated by an independent blinded re-coding across model families, and scales to the full adviser population; the validation codes and disagreement set, design-based inference, and survey weighting are provided so the measurement can be audited and extended.

Data Availability Statement

This study uses publicly available data (SEC Form ADV Part 1 structured data and Part 2A brochures via IAPD; SEC releases IA-6573 and IA-6574). The classification rubric, prompts, keyword-window rules, model snapshot and date, enforcement fingerprint, frozen sample, analysis and weighting code, the validation coding sheet, the independent cross-family codes, and the scoring code are deposited in a public repository (OSF/Zenodo) with a persistent identifier; the DOI will be provided on acceptance. Firm-level intermediate classifications are not released to avoid firm-level identification; only aggregate results are reported.

Author Contributions

SC conceived the study, built the data pipeline and classification instrument, conducted the analysis, and wrote the manuscript.

Conflict of Interest

The author declares that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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References

- Abis, S. (2022). Man vs. machine: quantitative and discretionary equity management. Working paper, Columbia Business School (SSRN 3717371).
- Acemoglu, D., Autor, D., Hazell, J., and Restrepo, P. (2022). Artificial intelligence and jobs: evidence from online vacancies. *J. Labor Econ.* 40, S293–S340.
- Babina, T., Fedyk, A., He, A. X., and Hodson, J. (2024). Artificial intelligence, firm growth, and product innovation. *J. Financ. Econ.* 151, 103745.
- Brynjolfsson, E., Li, D., and Raymond, L. (2025). Generative AI at work. *Q. J. Econ.* 140, 889–942.
- Byrt, T., Bishop, J., and Carlin, J. B. (1993). Bias, prevalence and kappa. *J. Clin. Epidemiol.* 46, 423–429.
- Cao, S., Jiang, W., Yang, B., and Zhang, A. L. (2024). How to talk when a machine is listening: corporate disclosure in the age of AI. *Rev. Financ. Stud.* 37, 3603–3642.
- Charles Schwab Advisor Services (2026). 2026 RIA and AI Research Study. San Francisco: Charles Schwab & Co.
- Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educ. Psychol. Meas.* 20, 37–46.
- Eisfeldt, A. L., Schubert, G., and Zhang, M. B. (2023). Generative AI and firm values. NBER Working Paper 31222.
- Gentzkow, M., Kelly, B., and Taddy, M. (2019). Text as data. *J. Econ. Lit.* 57, 535–574.
- Gilardi, F., Alizadeh, M., and Kubli, M. (2023). ChatGPT outperforms crowd workers for text-annotation tasks. *Proc. Natl. Acad. Sci. U.S.A.* 120, e2305016120.
- Grimmer, J., and Stewart, B. M. (2013). Text as data: the promise and pitfalls of automatic content analysis methods for political texts. *Political Anal.* 21, 267–297.
- Healy, P. M., and Palepu, K. G. (2001). Information asymmetry, corporate disclosure, and the capital markets: a review of the empirical disclosure literature. *J. Account. Econ.* 31, 405–440.
- Loughran, T., and McDonald, B. (2011). When is a liability not a liability? Textual analysis, dictionaries, and 10-Ks. *J. Finance* 66, 35–65.
- Loughran, T., and McDonald, B. (2016). Textual analysis in accounting and finance: a survey. *J. Account. Res.* 54, 1187–1230.
- Noy, S., and Zhang, W. (2023). Experimental evidence on the productivity effects of generative artificial intelligence. *Science* 381, 187–192.
- Rogers, E. M. (2003). *Diffusion of Innovations*, 5th Edn. New York: Free Press.
- U.S. Securities and Exchange Commission (2019). Commission Interpretation Regarding Standard of Conduct for Investment Advisers. Release No. IA-5248.

- U.S. Securities and Exchange Commission (2024a). In re Delphia (USA) Inc. Investment Advisers Act Release No. 6573, March 18, 2024.
- U.S. Securities and Exchange Commission (2024b). In re Global Predictions Inc. Investment Advisers Act Release No. 6574, March 18, 2024.
- U.S. Securities and Exchange Commission, Division of Examinations (2025). Fiscal Year 2026 Examination Priorities. Washington, DC: SEC.
- Wilson, E. B. (1927). Probable inference, the law of succession, and statistical inference. *J. Am. Stat. Assoc.* 22, 209–212.
- Ziems, C., Held, W., Shaikh, O., Chen, J., Zhang, Z., and Yang, D. (2024). Can large language models transform computational social science? *Comput. Linguist.* 50, 237–291.